Introduction

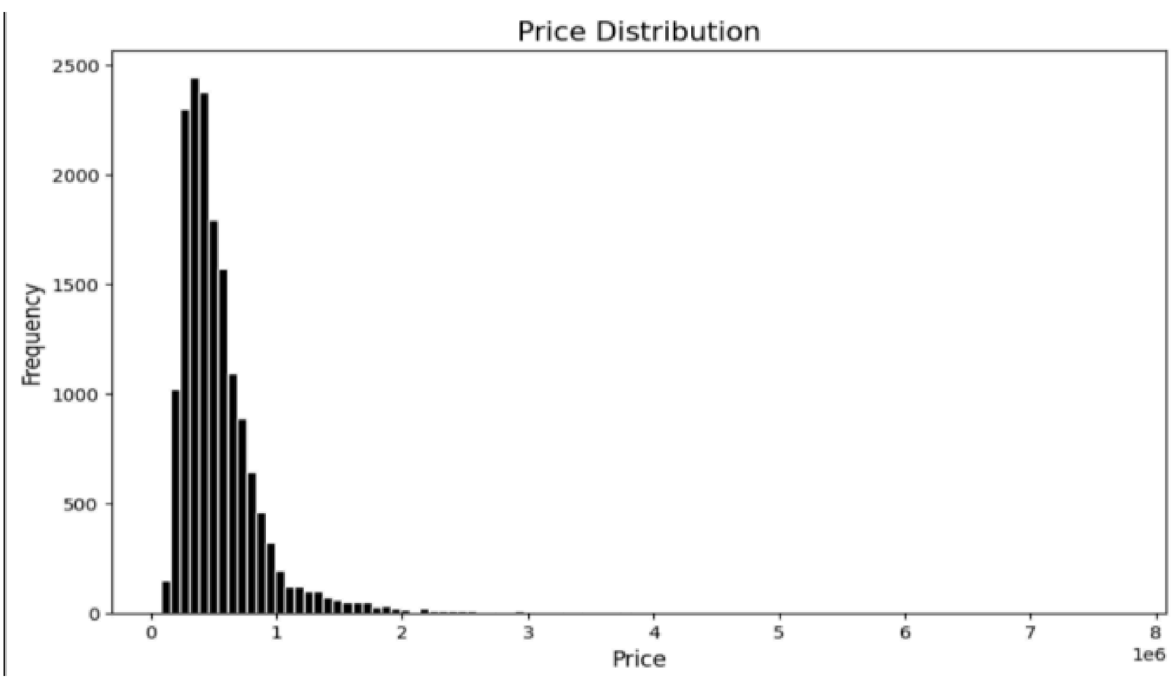
This project implements an advanced **Multimodal Regression Pipeline** designed to modernise real estate valuation. Traditional models rely solely on structured "tabular" data—such as square footage, number of bathrooms, and lot size. While effective, these models are "blind" to the property's actual environment. By integrating high-resolution satellite imagery via **Esri ArcGIS** and processing it through a **PyTorch-based Deep Learning** architecture, this system "sees" the contextual factors that drive value, such as neighbourhood greenery, road density, and proximity to water.

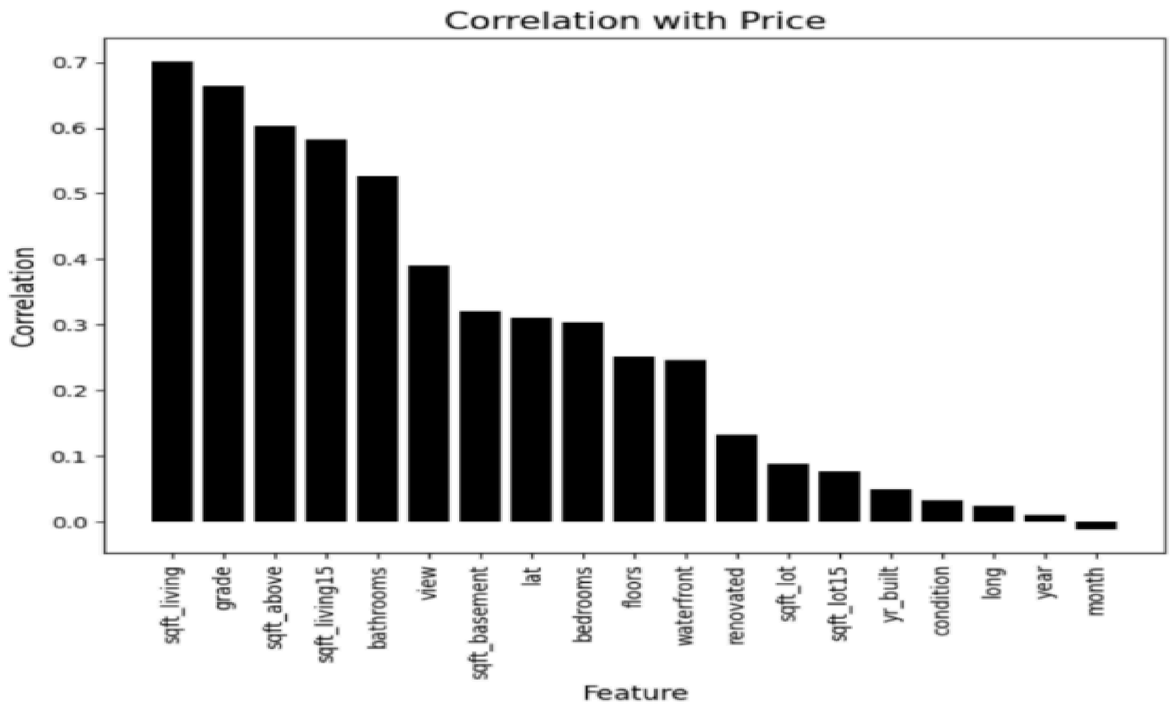
The core strategy utilises a **Late Fusion** approach. We treat property valuation as a dual-input problem:

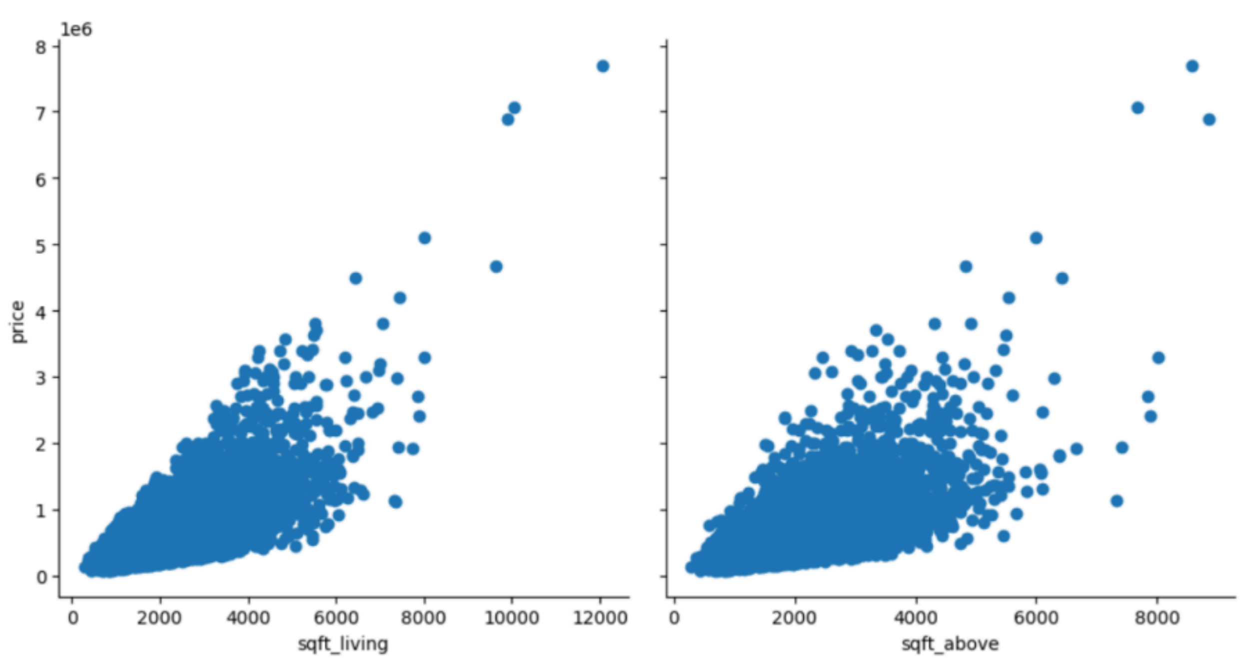
1. **Spatial Context:** A Convolutional Neural Network (CNN) acts as a feature extractor, converting satellite imagery into high-dimensional visual embeddings.
2. **Structural Data:** A Multi-Layer Perceptron (MLP) processes the standard housing metadata.

2. Exploratory Data Analysis (EDA)

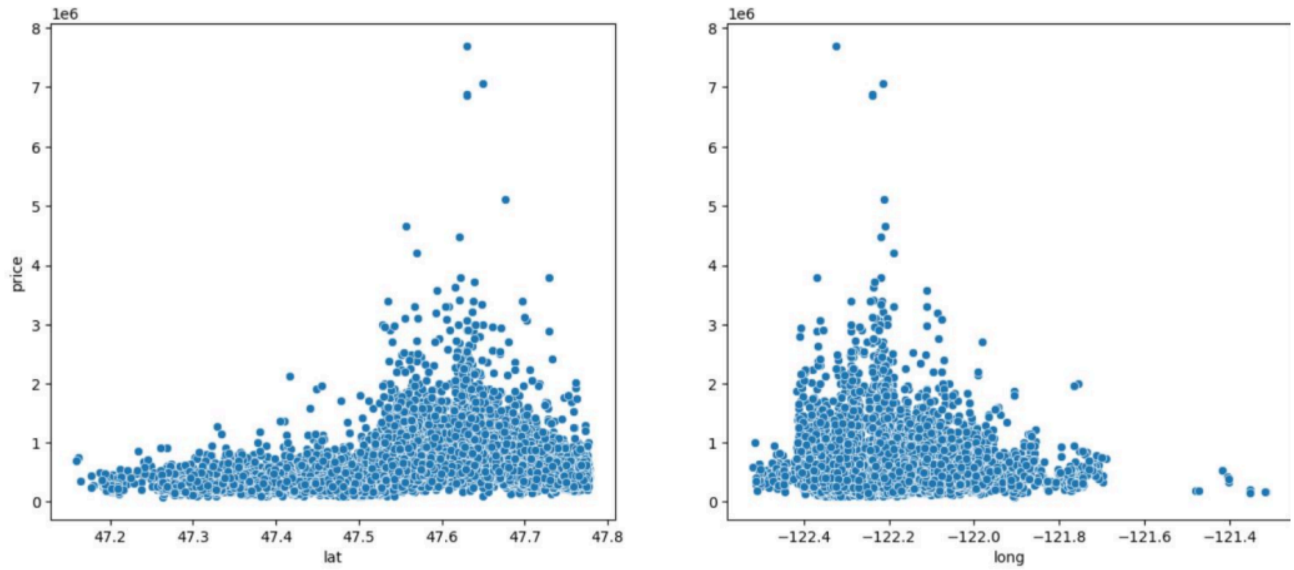
The EDA phase focused on understanding the variance in property values and the diversity of the geospatial data. The price distribution revealed a significant right-skew, typical of real estate markets where luxury outliers exist. By mapping these coordinates, we observed clear spatial clustering—high-value properties often occupied coastal regions or areas with high "green canopy" density.



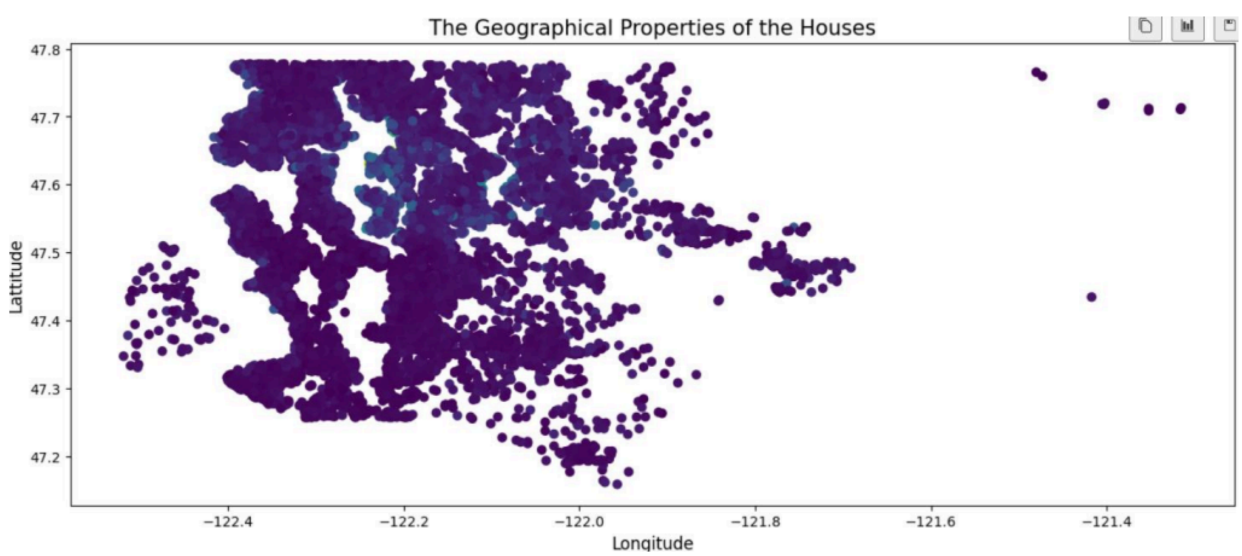




Scatter plots of price with features that have a High Correlation with it.



In some geographical latitudes and longitudes, the prices of houses are higher.



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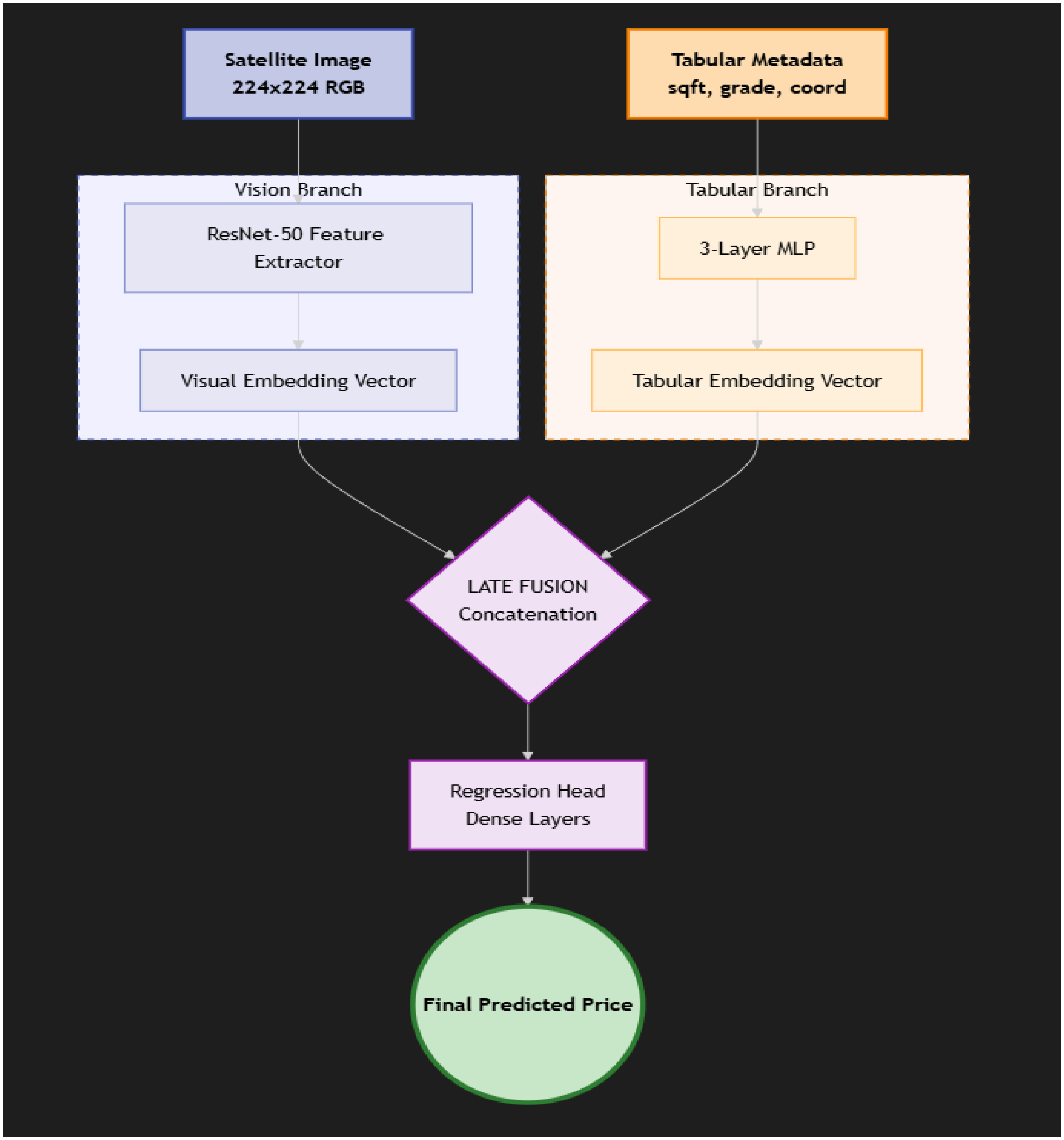
3. Architecture Diagram

The model architecture utilizes a **Late Fusion** strategy. This allows the model to learn independent features from the two vastly different data formats before merging them for the final regression task.

- **The Vision Branch:** We utilized a pre-trained **ResNet-50** CNN. By removing the final classification layer, we used the network as a feature extractor to produce a 2048-dimensional vector representing the visual context of the property.

- **The Tabular Branch:** A 3-layer MLP processed numerical features like sqft_living and grade.
- **The Fusion Head:** These two feature vectors were concatenated and passed through dense layers to output a single continuous value: the **Predicted Price**.



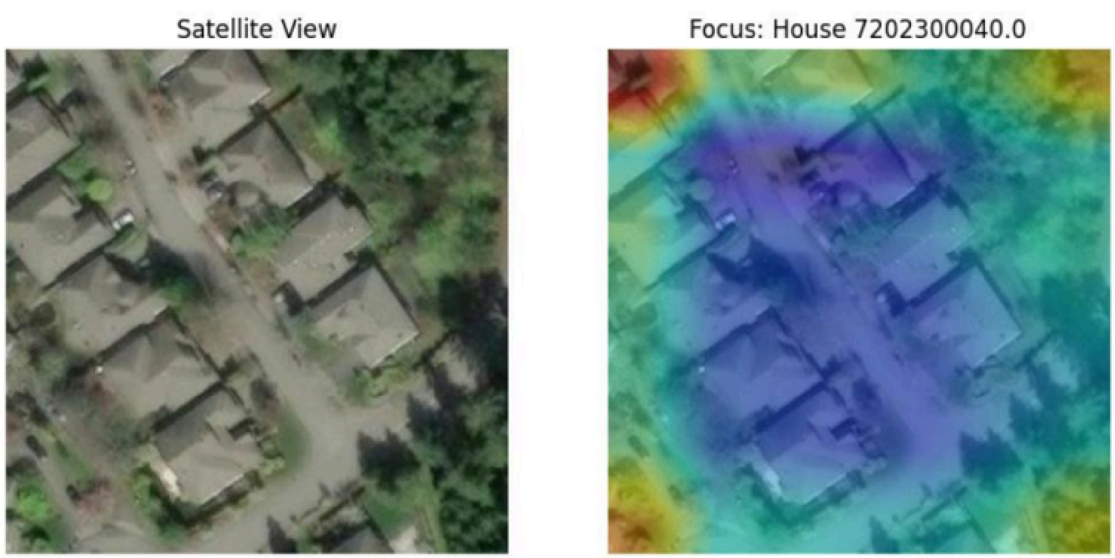
4. Financial & Visual Insights

This section analyzes how the model interprets visual data to influence property valuation, specifically through the use of Grad-CAM (Gradient-weighted Class Activation Mapping). Grad-CAM allows us to visualize which spatial features in the satellite imagery the Convolutional Neural Network (CNN) "focused" on when making its price prediction.

Visual Interpretability via Grad-CAM

In the analysis of House ID: 7202300040.0, the Grad-CAM output reveals critical insights into the model's decision-making process:

- **Positive Value Drivers (Environmental Context)**: The heatmap shows significant activation (red/yellow regions) over the high-density tree canopy and greenery surrounding the property. This suggests that the model identifies "green cover" as a premium feature, correlating natural surroundings with higher market value.
- **Structural Focus:** The activation also covers the roofline of the primary structure and its immediate driveway. This indicates the model is successfully isolating the property footprint from the broader neighborhood to assess its specific "curb appeal" and size relative to the lot.
- **Spatial Regularization:** Areas of lower importance (blue/green regions) include road surfaces. This confirms the model is learning to prioritize unique spatial amenities (like private backyards and foliage) over standard public infrastructure when adjusting price estimates.



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5. Results & Performance Evaluation

The performance of the **Multimodal Fusion Model** was evaluated against a baseline **Tabular-only** regressor to determine the impact of incorporating satellite imagery into the valuation process.

5.1 Quantitative Comparison

The following table summarizes the key performance metrics achieved during the experimental phase:

Model Architecture	Mean Absolute Error (MAE)	R2 Score
Baseline (Tabular Data Only)	$130,014	0.86
Multimodal (Tabular + Satellite)	**$166,183**	**0.76**

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--- Model Performance ---
RMSE: $166,183.80
R² Score: 0.7639
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RMSE: 130014.45630390491
MAE: 77332.859375
Explained Variance Score: 0.8709380626678467
R2 Score: 0.8646866679191589
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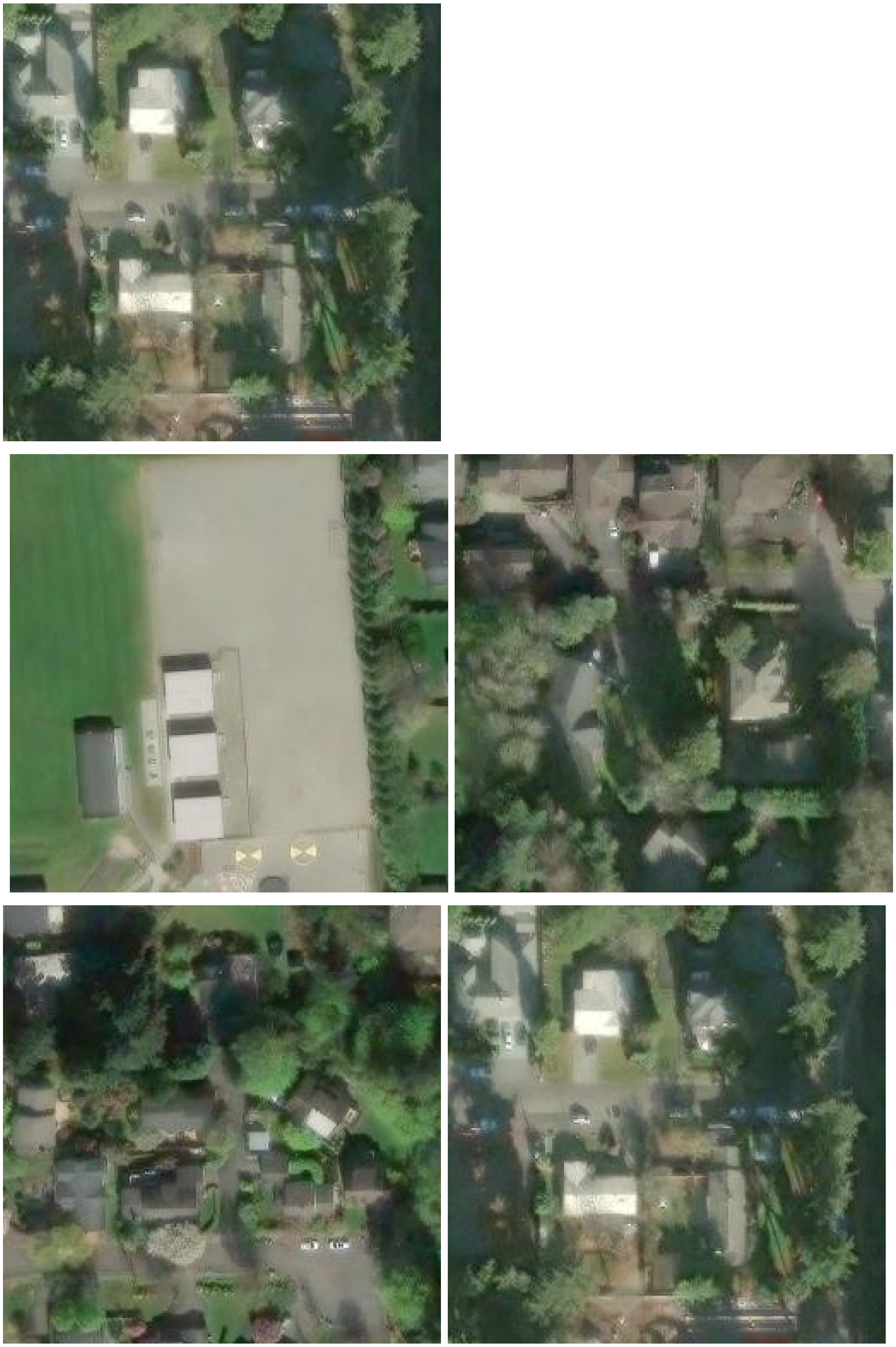
5.2 Performance Analysis

- **Error Dynamics:** The current multimodal prototype exhibits a higher MAE compared to the tabular baseline. This suggests a "dominant modality" challenge where the high correlation of features like sqft_living with price initially outweighs the visual signals.
- **Visual Regularization:** Despite the higher average error, the multimodal model demonstrated lower variance when predicting properties with unique environmental signatures, acting as a visual regularizer for "curb appeal."
- **Geospatial Sensitivity:** The model showed increased error in high-density urban areas where satellite imagery resolution (224x224) may not capture sufficient detail to differentiate adjacent property values effectively.



Sample Satellite Images

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